Django Complete Tutorial Notes

Part 1–8 + Reusable Apps + First Contribution

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1. Django Introduction

What is Django?

Django is a high-level Python web framework used to build secure, maintainable, and scalable web applications.

It includes tools for database operations, URL routing, templates, forms, authentication, and administration.

Main Features

- Built-in authentication and authorization
- Object-Relational Mapper (ORM)
- Automatic administration interface
- URL routing
- Template engine
- Form validation
- Security protections
- Automated testing support
- Reusable application architecture

What is MVT?

Django follows the Model-View-Template (MVT) architectural pattern.

Component	Responsibility
Model	Defines data and database interactions
View	Processes requests and prepares responses
Template	Presents information to users

Request Flow

Browser → URL Configuration → View → Model (if needed) → Template (if needed) → HTTP Response

Django vs Flask

Django	Flask
Full-featured framework	Lightweight framework
Built-in ORM and admin	Extensions commonly provide additional features
Structured application conventions	Flexible application structure
Suitable for many full-stack web applications	Suitable for small services and customized architectures

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2. Part 1 — Project, App, Views and URLs

Step 1: Create a Virtual Environment

```
python -m venv .venv
```

Activate it on Windows:

```
.venv\Scripts\activate
```

Activate it on Linux or macOS:

```
source .venv/bin/activate
```

Step 2: Install Django

```
python -m pip install django
```

Check the installed version:

```
python -m django --version
```

Step 3: Create a Project

```
django-admin startproject mysite
cd mysite
```

Step 4: Run the Development Server

```
python manage.py runserver
```

Open:
`http://127.0.0.1:8000/`

Step 5: Create an Application

```
python manage.py startapp polls
```

A Django project can contain multiple applications.

Example structure:

```
mysite/
├── manage.py
├── mysite/
│   ├── settings.py
│   ├── urls.py
│   ├── asgi.py
│   └── wsgi.py
└── polls/
    ├── admin.py
    ├── apps.py
    ├── models.py
    ├── tests.py
    └── views.py
```

Step 6: Register the Application

In `mysite/settings.py`:

```
INSTALLED_APPS = [
    "django.contrib.admin",
    "django.contrib.auth",
    "django.contrib.contenttypes",
    "django.contrib.sessions",
    "django.contrib.messages",
    "django.contrib.staticfiles",
    "polls",
]
```

Step 7: Create a Function-Based View

File: `polls/views.py`

```
from django.http import HttpResponse


def home(request):
    return HttpResponse("Welcome to Django!")
```

Step 8: Configure Application URLs

Create `polls/urls.py`:

```
from django.urls import path
from . import views

app_name = "polls"

urlpatterns = [
    path("", views.home, name="home"),
]
```

Step 9: Connect Application URLs

File: `mysite/urls.py`

```
from django.contrib import admin
from django.urls import include, path

urlpatterns = [
    path("admin/", admin.site.urls),
    path("polls/", include("polls.urls")),
]
```

Visit:
`http://127.0.0.1:8000/polls/`

Important Concepts

- `request`: Contains incoming HTTP request information.
- `HttpResponse`: Returns an HTTP response.
- `path()`: Maps a URL pattern to a view.

- `include()` : Includes another URL configuration.
- `app_name` : Defines an application namespace.
- `name` : Defines a URL pattern's name.

---

3. Part 2 — Database, Models, ORM and Admin

What is a Model?

A Django model is a Python class that defines data structure and database behavior.

File: `polls/models.py`

```
from django.db import models


class Question(models.Model):
    question_text = models.CharField(max_length=200)
    pub_date = models.DateTimeField()

    def __str__(self):
        return self.question_text
```

Common Model Fields

Field	Purpose
`CharField`	Short text
`TextField`	Long text
`IntegerField`	Integer values
`PositiveIntegerField`	Non-negative integer values
`BooleanField`	True or false
`DateTimeField`	Date and time
`DateField`	Date
`EmailField`	Email-formatted text
`ForeignKey`	Many-to-one relationship

`OneToOneField`	One-to-one relationship
`ManyToManyField`	Many-to-many relationship

Model Relationships

ForeignKey

```
class Choice(models.Model):
    question = models.ForeignKey(
        Question,
        on_delete=models.CASCADE,
        related_name="choices",
    )
    choice_text = models.CharField(max_length=200)
    votes = models.PositiveIntegerField(default=0)
```

One question can have multiple choices.

```
question.choices.all()
```

The `related_name="choices"` setting defines the reverse relationship.

OneToOneField

Useful when one model should have exactly one associated record.

```
from django.conf import settings
from django.db import models


class UserProfile(models.Model):
    user = models.OneToOneField(
        settings.AUTH_USER_MODEL,
        on_delete=models.CASCADE,
    )
    bio = models.TextField(blank=True)
```

ManyToManyField

```
class Student(models.Model):
    name = models.CharField(max_length=100)


class Course(models.Model):
    title = models.CharField(max_length=100)
    students = models.ManyToManyField(Student, related_name="courses")
```

A course can have multiple students, and a student can enroll in multiple courses.

Migrations

Create migration files:

```
python manage.py makemigrations
```

Apply migrations:

```
python manage.py migrate
```

Check migration status:

```
python manage.py showmigrations
```

Migrations keep the database schema synchronized with model changes.

Django ORM

The ORM allows database operations through Python objects.

Create

```
Question.objects.create(
    question_text="What is Django?",
    pub_date=timezone.now(),
)
```

Add this import when using the example:

```
from django.utils import timezone
```

Retrieve All Objects

```
Question.objects.all()
```

Retrieve One Object

```
question = Question.objects.get(pk=1)
```

`get()` raises `Question.DoesNotExist` if no matching object exists.

Filter

```
Question.objects.filter(question_text__icontains="django")
```

Order By

```
Question.objects.order_by("-pub_date")
```

Update

```
Question.objects.filter(pk=1).update(
    question_text="Updated question"
)
```

Delete

```
Question.objects.filter(pk=1).delete()
```

Count

```
Question.objects.count()
```

Useful ORM Lookups

- `exact`
- `contains`
- `icontains`
- `gt`
- `gte`
- `lt`
- `lte`
- `in`
- `isnull`

Example:

```
Question.objects.filter(pk__gt=5)
```

Register Models in Django Admin

File: `polls/admin.py`

```
from django.contrib import admin
from .models import Choice, Question


class ChoiceInline(admin.TabularInline):
    model = Choice
    extra = 2


@admin.register(Question)
class QuestionAdmin(admin.ModelAdmin):
    list_display = ("question_text", "pub_date")
    search_fields = ("question_text",)
    inlines = [ChoiceInline]


@admin.register(Choice)
class ChoiceAdmin(admin.ModelAdmin):
    list_display = ("choice_text", "question", "votes")
    list_filter = ("question",)
```

Create an admin account:

```
python manage.py createsuperuser
```

Open:
`http://127.0.0.1:8000/admin/`

---

4. Part 3 — Views and Templates

Function-Based Views

```
from django.shortcuts import render


def home(request):
    return render(request, "home.html")
```

`render()` combines a template with context data and returns an HTTP response.

Passing Context to a Template

```
def home(request):
    context = {
        "name": "Faruk",
        "role": "Backend Developer",
    }

    return render(request, "home.html", context)
```

Template:

```
<h1>Hello, {{ name }}</h1>
<p>Role: {{ role }}</p>
```

Template Variables

```
{{ question.question_text }}
```

Template Tags

If

```
{% if questions %}
    <p>Questions are available.</p>
{% else %}
    <p>No questions found.</p>
{% endif %}
```

For

```
{% for question in questions %}
    <p>{{ question.question_text }}</p>
{% empty %}
    <p>No questions available.</p>
{% endfor %}
```

URL Tag

```
<a href="{% url 'polls:home' %}">Home</a>
```

Named URLs reduce hard-coded URL strings.

Dynamic URLs

In `polls/urls.py`:

```
path("<int:pk>/", views.detail, name="detail")
```

In `polls/views.py`:

```
from django.shortcuts import get_object_or_404, render


def detail(request, pk):
    question = get_object_or_404(Question, pk=pk)

    return render(
        request,
        "polls/detail.html",
        {"question": question},
    )
```

The URL parameter name `pk` must match the view argument `pk`.

Static Templates and Template Directories

Recommended application template structure:

```
polls/
└── templates/
    └── polls/
        ├── index.html
        ├── detail.html
        └── results.html
```

Using an application-specific directory helps prevent template-name collisions.

---

5. Part 4 — Forms, Voting and Generic Views

Django Forms

Forms validate submitted input and provide error messages.

File: `polls/forms.py`

```
from django import forms


class ContactForm(forms.Form):
    name = forms.CharField(max_length=100)
    email = forms.EmailField()
    message = forms.CharField(widget=forms.Textarea)
```

Handling a Form

```
from django.shortcuts import render
from .forms import ContactForm


def contact(request):
    if request.method == "POST":
        form = ContactForm(request.POST)

        if form.is_valid():
            name = form.cleaned_data["name"]
            email = form.cleaned_data["email"]

            # Process the validated data here.
            return render(
                request,
                "contact_success.html",
                {"name": name, "email": email},
            )
    else:
        form = ContactForm()

    return render(
        request,
        "contact.html",
        {"form": form},
    )
```

Template:

```
<form method="post">
    {% csrf_token %}
    {{ form.as_p }}
    <button type="submit">Submit</button>
</form>
```

Always validate submitted data on the server. Client-side validation alone is not sufficient.

ModelForm

A `ModelForm` creates a form based on a Django model.

```
from django import forms
from .models import Question


class QuestionForm(forms.ModelForm):
    class Meta:
        model = Question
        fields = ["question_text"]
```

Use an explicit list of fields rather than exposing every model field to users.

Class-Based Views

```
from django.http import HttpResponse
from django.views import View


class HomeView(View):
    def get(self, request):
        return HttpResponse("Home Page")

    def post(self, request):
        return HttpResponse("Form submitted")
```

URL:

```
path("", HomeView.as_view(), name="home")
```

Generic CBVs

ListView

```
from django.views.generic import ListView
from .models import Question


class QuestionListView(ListView):
    model = Question
    template_name = "polls/index.html"
    context_object_name = "questions"
    ordering = ["-pub_date"]
    paginate_by = 10
```

DetailView

```
from django.views.generic import DetailView


class QuestionDetailView(DetailView):
    model = Question
    template_name = "polls/detail.html"
    context_object_name = "question"
```

CreateView

```
from django.urls import reverse_lazy
from django.views.generic import CreateView


class QuestionCreateView(CreateView):
    model = Question
    fields = ["question_text"]
```

```
    template_name = "polls/question_form.html"
    success_url = reverse_lazy("polls:index")
```

UpdateView

```
from django.views.generic import UpdateView


class QuestionUpdateView(UpdateView):
    model = Question
    fields = ["question_text"]
    template_name = "polls/question_form.html"
    success_url = reverse_lazy("polls:index")
```

DeleteView

```
from django.views.generic import DeleteView


class QuestionDeleteView(DeleteView):
    model = Question
    template_name = "polls/question_confirm_delete.html"
    success_url = reverse_lazy("polls:index")
```

Deletion should be performed through POST after confirmation.

Voting with `F()` Expressions

```
from django.db.models import F

Choice.objects.filter(
    pk=choice.pk,
    question=question,
).update(votes=F("votes") + 1)
```

`F()` allows the database to increment the current value and helps avoid lost updates.
However, `F()` does not prevent duplicate voting. If only one vote per user is allowed, store voting records and enforce uniqueness with a database constraint.

---

6. Part 5 — Automated Testing

Django provides a test framework built on Python's `unittest`.
File: `polls/tests.py`

```
from django.test import TestCase
from django.urls import reverse

from .models import Choice, Question


class PollsTests(TestCase):
```

```
    def setUp(self):
        self.question = Question.objects.create(
            question_text="What is your favorite language?"
        )

        self.choice = Choice.objects.create(
            question=self.question,
            choice_text="Python",
        )

    def test_index_returns_200(self):
        response = self.client.get(
            reverse("polls:index")
        )

        self.assertEqual(response.status_code, 200)

    def test_index_contains_question(self):
        response = self.client.get(
            reverse("polls:index")
        )

        self.assertContains(
            response,
            "What is your favorite language?",
        )

    def test_missing_question_returns_404(self):
        response = self.client.get(
            reverse("polls:detail", args=[999999])
        )

        self.assertEqual(response.status_code, 404)
```

Run tests:

```
python manage.py test
```

Run tests for one application:

```
python manage.py test polls
```

What Should Be Tested?

- HTTP status codes
- Template rendering
- Model behavior
- Form validation
- Successful form submission
- Invalid input

- Authentication and permissions
- Voting behavior
- Duplicate-vote prevention, if required
- Database constraints

Tests help detect regressions when the application changes.

---

7. Part 6 — Static Files

Static files include CSS, JavaScript, and images that are served separately from dynamic application responses.

Recommended Structure

```
polls/
├── static/
│   └── polls/
│       ├── css/
│       │   └── style.css
│       ├── js/
│       │   └── app.js
│       └── images/
│           └── logo.png
└── templates/
    └── polls/
        └── index.html
```

Configure Static Files

In `settings.py`:

```
STATIC_URL = "static/"
```

For production collection, configure an output directory:

```
STATIC_ROOT = BASE_DIR / "staticfiles"
```

Load Static Files in a Template

```
{% load static %}

<!DOCTYPE html>
<html lang="en">
<head>
    <link rel="stylesheet" href="{% static 'polls/css/style.css' %}">
</head>
<body>
    <h1>Django Application</h1>

    <script src="{% static 'polls/js/app.js' %}"></script>
```

```
</body>
</html>
```

Example CSS

File: `polls/static/polls/css/style.css`

```
body {
    font-family: Arial, sans-serif;
    margin: 30px;
}

h1 {
    color: #2457a7;
}
```

Collect Static Files

```
python manage.py collectstatic
```

In production, configure a suitable static-file server or storage backend. Do not rely on Django's development static-file serving in production.

---

8. Part 7 — Django Admin Customization

Django Admin allows authorized staff to manage application data.

Basic Registration

```
from django.contrib import admin
from .models import Question

admin.site.register(Question)
```

Customize List Display

```
@admin.register(Question)
class QuestionAdmin(admin.ModelAdmin):
    list_display = ("question_text", "pub_date")
```

Search

```
search_fields = ("question_text",)
```

Filtering

```
list_filter = ("pub_date",)
```

Ordering

```
ordering = ("-pub_date",)
```

Inline Models

```
class ChoiceInline(admin.TabularInline):
    model = Choice
    extra = 2
```

Then add:

```
inlines = [ChoiceInline]
```

Read-Only Fields

```
readonly_fields = ("pub_date",)
```

Read-only fields can prevent ordinary editing through the admin form. They do not replace application-level security or database integrity controls.

Admin Best Practices

- Register models intentionally.
- Use `list_display` for useful summaries.
- Add appropriate search and filter fields.
- Restrict admin access to authorized staff.
- Avoid exposing sensitive fields unnecessarily.
- Use read-only fields for data that should not be edited through admin forms.

---

9. Part 8 — Third-Party Packages and Debug Toolbar

Install a Third-Party Package

Example:

```
python -m pip install django-debug-toolbar
```

Record dependencies in a requirements file:

```
python -m pip freeze > requirements.txt
```

For a team project, consider a deliberately maintained dependency file or lockfile rather than relying only on an uncontrolled snapshot of the environment.

Configure Django Debug Toolbar

Step 1: Install

```
python -m pip install django-debug-toolbar
```

Step 2: Add to Installed Apps

In `settings.py`:

```
INSTALLED_APPS = [
    # Existing Django apps...
```

```
    "debug_toolbar",
]
```

Step 3: Add Middleware

Add the middleware near the beginning of `MIDDLEWARE`, after any middleware that must precede it according to the package documentation.

```
MIDDLEWARE = [
    "django.middleware.security.SecurityMiddleware",
    "debug_toolbar.middleware.DebugToolbarMiddleware",
    # Remaining middleware...
]
```

Use the package's installation documentation to verify the appropriate middleware order for your project.

Step 4: Configure Internal IPs

For local development:

```
INTERNAL_IPS = [
    "127.0.0.1",
]
```

Step 5: Add Toolbar URLs

In `mysite/urls.py`:

```
from django.conf import settings
from django.urls import include, path

urlpatterns = [
    path("polls/", include("polls.urls")),
]

if settings.DEBUG:
    import debug_toolbar

    urlpatterns += [
        path("__debug__/", include(debug_toolbar.urls)),
    ]
```

Preserve your existing admin route when applying this example.

What Does Debug Toolbar Provide?

- SQL query inspection
- Request and response information
- Template rendering details
- Cache information
- Timing and performance diagnostics

Security warning: Debug Toolbar should be used only in controlled development environments. Do not expose it publicly in production.

---

10. Reusable Django Apps and Python Packaging

A reusable Django app is designed to be installed and used in multiple Django projects.

Examples include:

- Reusable authentication components
- Notification systems
- Payment integrations
- Reusable model utilities
- Common API components

Example App Structure

```
django-example-app/
├── pyproject.toml
├── README.md
├── LICENSE
├── example_app/
│   ├── __init__.py
│   ├── apps.py
│   ├── models.py
│   ├── views.py
│   ├── urls.py
│   └── migrations/
│       └── __init__.py
└── tests/
```

Create an App Configuration

File: `example_app/apps.py`

```
from django.apps import AppConfig


class ExampleAppConfig(AppConfig):
    default_auto_field = "django.db.models.BigAutoField"
    name = "example_app"
```

Example `pyproject.toml`

```
[build-system]
requires = ["setuptools>=68"]
build-backend = "setuptools.build_meta"

[project]
name = "django-example-app"
version = "0.1.0"
```

```
description = "A reusable Django application"
requires-python = ">=3.10"
dependencies = [
    "Django>=5.2,<6.0",
]

[tool.setuptools.packages.find]
include = ["example_app*"]
```

Choose supported Python and Django versions appropriate to the project before publishing. The version ranges above are illustrative, not a guarantee of compatibility with every environment.

Install the Package Locally

From the package root:

```
python -m pip install -e .
```

Editable installation makes development changes available without reinstalling the package after every source edit.

Register the App

In the consuming project's `settings.py`:

```
INSTALLED_APPS = [
    # Existing applications...
    "example_app",
]
```

If the app provides URLs, include them in the project URL configuration:

```
path("example/", include("example_app.urls"))
```

Packaging Best Practices

- Include a README with installation and usage instructions.
- Include a suitable open-source license.
- Declare runtime dependencies.
- Include migrations when the app has database models.
- Write tests.
- Avoid hard-coded project-specific settings.
- Document required settings and URL configuration.
- Build and test the package before publishing.

---

11. First Contribution to Django

Contributing to Django or another open-source project requires a reproducible development environment, focused changes, and tests.

Step 1: Understand the Project

Read the official contribution guide and code of conduct.

Django documentation:

https://docs.djangoproject.com/

Django contribution documentation:

https://docs.djangoproject.com/en/stable/internals/contributing/

Step 2: Fork the Repository

On GitHub, fork the Django repository:

https://github.com/django/django

Step 3: Clone Your Fork

Replace `YOUR_USERNAME` with your GitHub username.

```
git clone https://github.com/YOUR_USERNAME/django.git
cd django
```

Add the official repository as the upstream remote:

```
git remote add upstream https://github.com/django/django.git
```

Step 4: Create a Virtual Environment

```
python -m venv .venv
```

Activate it using the command appropriate for your operating system.

Step 5: Install Development Dependencies

Follow Django's current contributor instructions for installing development requirements. The exact commands can vary by branch and contributor setup.

Step 6: Create a Feature Branch

```
git switch -c fix-small-bug
```

Use a descriptive branch name related to the issue.

Step 7: Choose an Issue

Look for a clearly described, reproducible bug or documentation improvement that is appropriate for a first contribution.

Before starting, check whether the issue is already being addressed and follow the project's contribution guidelines.

Step 8: Reproduce the Problem

Write a minimal example or test that demonstrates the problem.

Reproducing the issue helps ensure that the fix addresses the actual behavior rather than an assumption.

Step 9: Make a Small Change

Prefer focused changes with a clear purpose.

Follow the project's formatting, naming, testing, and documentation conventions.

Step 10: Run Tests

Run the relevant tests using the project's documented testing instructions.

A change that fixes a bug should normally include a regression test when practical.

Step 11: Review Your Changes

```
git diff
git status
```

Check for unrelated changes, missing tests, formatting problems, and unintended behavior.

Step 12: Commit Your Changes

```
git add .
git commit -m "Fix a small issue"
```

Use a concise, descriptive commit message consistent with the project's conventions.

Step 13: Push Your Branch

```
git push origin fix-small-bug
```

Step 14: Open a Pull Request

Open a pull request from your fork and branch to the appropriate upstream branch.

Include:

- A description of the problem
- The cause of the issue
- The change you made
- Tests performed
- Related issue references, if applicable

Respond constructively to review feedback and update your branch as requested.

---

12. Important Commands and Practice Project

Essential Django Commands

Command	Purpose
`django-admin startproject mysite`	Create a project
`python manage.p`	Create an app

`y startapp polls`	
`python manage.py runserver`	Start development server
`python manage.py check`	Check project configuration
`python manage.py makemigrations`	Create migrations
`python manage.py migrate`	Apply migrations
`python manage.py showmigrations`	Display migration status
`python manage.py createsuperuser`	Create admin account
`python manage.py shell`	Open Django shell
`python manage.py test`	Run automated tests
`python manage.py collectstatic`	Collect static files

Practice Project: Blog Application

Build a complete Blog application using the concepts in these notes.

Core Features

1. List blog posts.
2. View individual posts.
3. Create posts.
4. Update posts.
5. Delete posts with confirmation.
6. Add categories and tags.
7. Implement search.
8. Add pagination.
9. Add authentication.
10. Restrict editing to authorized users.
11. Upload and display images.
12. Write automated tests.
13. Customize Django Admin.
14. Deploy using production-appropriate settings.

Suggested Models

- `Post`
- `Category`
- `Tag`
- `Comment`

Suggested Learning Order

1. Build models and migrations.
2. Register models in Django Admin.
3. Implement `ListView` and `DetailView`.
4. Add create, update, and delete operations.
5. Add forms and validation.
6. Implement authentication and permissions.
7. Add static files and media handling.
8. Write automated tests.
9. Optimize database queries.
10. Prepare the application for deployment.

Final Revision Checklist

- Understand Django's MVT architecture.
- Create projects and applications.
- Configure URL routing.
- Write function-based and class-based views.
- Create models and migrations.

- Practice ORM filtering and relationships.
- Build templates and forms.
- Use generic CBVs for CRUD operations.
- Customize Django Admin.
- Configure static files.
- Install and configure development packages.
- Write automated tests.
- Package a reusable Django app.
- Understand Git and GitHub contribution workflows.
- Build and test a complete Blog application.

Final Goal: Build a maintainable Django application, validate user input, protect application data, test important behavior, package reusable components, and contribute confidently to open-source projects.